APPLICATION EXAMPLE

Integration and Usage of the WinCC OA Text-To-Speech Application Example

SIMATIC WinCC Open Architecture

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Component	Open Source Software [Yes/No]	Acknowledgements/Comment	License conditions and copyright notices
say ^0.16.0	Yes	MIT	Marak/say.js: TTS (text to speech) for node.js. send text from node.js to your speakers.
heap-js ^2.7.1	Yes	BSD 3-Clause	ignlg/heap-js: Efficient Binary heap (priority queue, binary tree) data structure for JavaScript / TypeScript. Includes JavaScript methods, Python's heapq module methods, and Java's PriorityQueue methods.

Table 0-1 OSS Packages

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1. Introduction

1.1. Overview

This document describes the integration of the WinCC OA Text-to-Speech Manager within WinCC OA projects. The manager allows operators to receive audible notifications and announcements in real-time, to configure speech synthesis settings through dedicated panels, to manage message priorities and queuing automatically, and to integrate seamlessly with the alarm system.

The solution was tested with Windows 11 and English language

1.2. Components used

This application example has been created with the following hard- and software components:

Component	Number	Article number
WinCC OA 3.20 Server Basis	1	6AV6355-1AA50-0BA0
WinCC OA V3.20, Para Standard	1	6AV6355-1AA50-0CH0

Table 1-1 WinCC OA Licenses

You can purchase these components from the Siemens Industry Mall.

1.3. Requirements

- Installed WinCC OA version 3.20
- Installed Node.js® version 22

This application example consists of the following components:

Component	File name	Note
Manual	WinCCOAtextToSpeechExample.pdf	This document
Application	T2S.zip	Panels, ctrl scripts, TS source code

Table 1-2 Application Example Parts

2. Engineering

2.1. Project integration

2.1.1. System requirements

- WinCC OA Version 3.20 P1 or higher installation and license (incl. Para)
- Minimal system requirements for WinCC OA (see WinCC OA Documentation)

2.1.2. Installation WinCC OA

To carry out the installation of WinCC OA, please follow the steps in the WinCC OA documentation.
You can find the documentation under the following link: Installation

2.1.3. Preparation

Unpack the supplied ZIP archive "T2S.zip" into the folder with a project where the Javascript text-to-speech manager is planned to be used.
The ZIP-archive contains 4 folders:

- dplist - Contains DP list for text-to-speech manager
- javascript – Contains source code for text-to-speech manager
- panels – Contains configuration/test panel
- scripts – Contains Control/Ctrl++ libs and classes

2.1.4. Creating a WinCC OA Project

To create your own project and use it with the manager, steps form online documentation Create project can be followed:

2.1.5. Configuration and usage of the manager.

Import the datapoint list texttospeech.dpl, (Import From ASCII File (Mass Engineering)) into your WinCC OA project. After successful import, the following new2 DpTypes will be created in your project

TextToSpeech DpType

The TextToSpeech DpType contains configuration and control datapoint elements for the text-to-speech engine

- **filter** (string) – contains JavaScript filter expression for filtering based on datapoint element name
- **dpes** (array) - contains array of filtered datapoint elements that match the filter
- **config** (string) - contains configuration JSON with TTS settings.

Config JSON Structure is:

{ "voice": "Microsoft David Desktop", "speed": 1.0, "maxQueueSize": 5, "language": "en-US", "availableVoices": ["Microsoft David Desktop", "Microsoft Zira Desktop"], "isInitialized": true }

Config Parameters:

- *voice* (string): Selected TTS voice name
- *speed* (float): Speech speed multiplier (0.1 - 3.0, default: 1.0 – default OS speed)
- *maxQueueSize* (int): Maximum number of queued messages
- *language* (string): Langage code (e.g., "en-US", "en-GB", "de-DE"), voices for language MUST be installed and available on your computer
- *availableVoices* (array): List of available system voices

TextToSpeechMessage DpType

The TextToSpeechMessage DpType is used to send messages to the TTS engine.

Message JSON Structure:

{ "managerNumber": 3, "dpe": "", "text": "This is a test message", "voice": "", "speed": 0, "priority": 5 }

Message Parameters:

- *managerNumber* (int): Target manager number
- *dpe* (*string*): Source datapoint element can be used for filtration)
- *text* (*string*): Message text to be spoken
- *voice* (string): Optional override voice for this specific message (empty = use config voice)
- *speed* (int): Optional override speed for this specific message (0 = use config speed, 0.1x-3.0x)
- *priority* (int): Message priority (1-maxint (9007199254740991), higher = more important)

Default Datapoints created after importing the datapoint list, the following datapoints are automatically created:

- **TTS_engine_3** (TextToSpeech) Main configuration datapoint for Text-To-speech manager #3:
- **Text2SpeechMsg (**TextToSpeechMessage**)** Message output datapoint:
- **alertVM** (_AlertClass) Alert class for voice message integration, it uses Control script to call sendMessage() function. **Alert class** is an internal DPE and does not appear in the PARA module by default. It will only be visible if the appropriate checkbox is enabled.

Naming Convention

If you want to define a new JS Manager with a new manager number but the same functionalities, a new datapoint must be created following this naming convention

Text-to-speech engine datapoints follow the naming pattern:

TTS_engine_<manager_number>

Examples**:**

TTS_engine_3 - Specific manager #3

TTS_engine_5 - Specific manager #5

Libraries and classes:

voiceMessageLib.ctl - Library for sending TTS messages, provides the sendMessage() function.
TSHandler.ctl - class for TTS configuration management, provides configuration management class with methods for reading/writing TTS configuration.

1. To install dependencies and transpile source code following terminal commands should be executed in the folder <ProjectName>\javascript\text-2-speech

   - npm install
   - npx tsc

```
PS C:\WinCC_OA_Proj\texttospeech\javascript\text-2-speech> npm install

  up to date, audited 9 packages in 3s

  found 0 vulnerabilities
PS C:\WinCC_OA_Proj\texttospeech\javascript\text-2-speech> npx tsc
PS C:\WinCC_OA_Proj\texttospeech\javascript\text-2-speech>
```

2. After source codes are transpiled new JavaScript manger should be added into project console with following options: -num 3 text-2-speech/dist/index.js

3. Configuration panel tts_config.xml provides a graphical interface for:

   Configuring voice settings – voice for the speech, which are available for the current OS, more voices can be installed in your OS.

   !!!!!!!!!Voice and Language arrays are not relaying on each other, what means if user select a language, the will- see all voices which are available on the computer.

   Adjusting speech speed – speed of the speech

   Setting queue size – maximum queued messages

   Managing datapoint filter. Filter is a JavaScript expression used as the body of a filter function with the signature name => <your expression>. The expression receives the datapoint element name and should evaluate to true to allow it through, or false to block it. A DPE will be announced only if it contains a non-empty value and passes this filter.

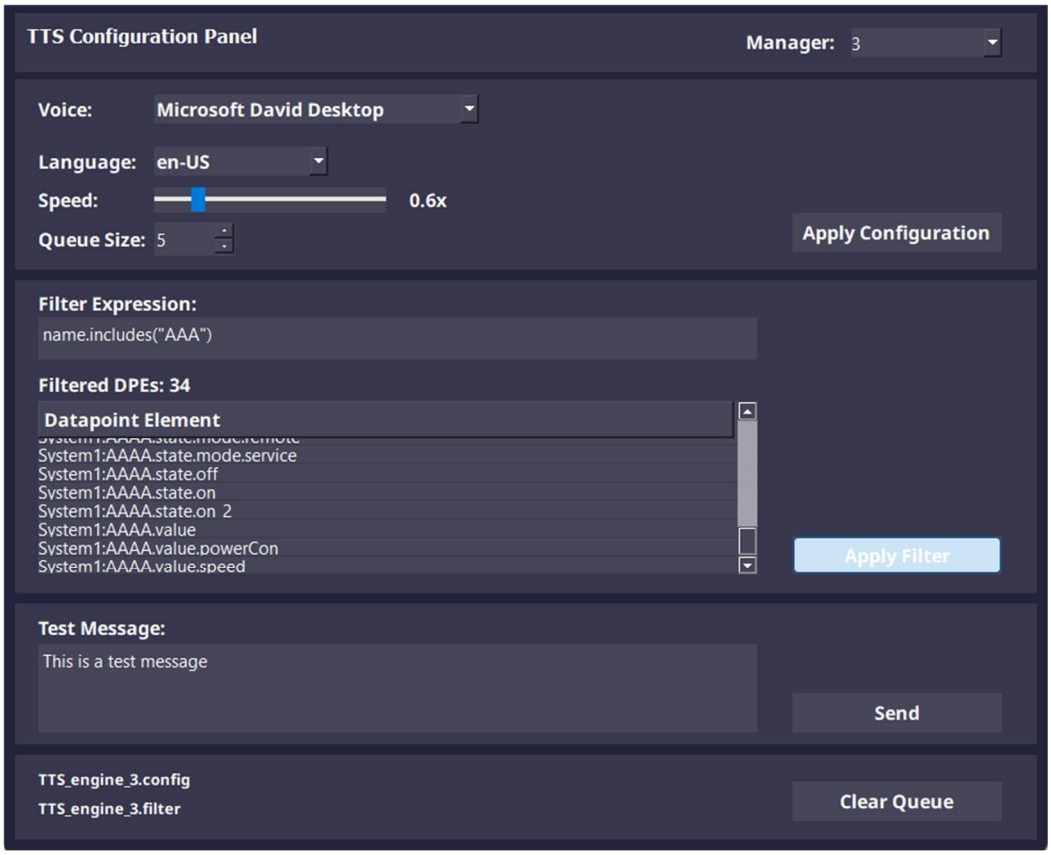
   Examples:

   - true – allow all DPEs
   - name.includes('Motor') – allow only DPEs whose name contains Motor
   - name.startsWith('System') – allow only DPEs starting with System
   - { if (name.includes('Alarm')) return true; return false; } – block statement form

   Any truthy value (non-zero number, non-empty string) passes the filter; false, 0, or null blocks it. For more details, see Iterator.prototype.filter() - JavaScript | MDN

   Testing messages – it is possible to send test messages; test messages are sent to chosen manager.

   Clear message queue - removes all pending messages currently queued for a selected manager.



4. Alert class alertVM contains an example, how alerts for speech can be configured. For testing purposes this class can be used as alert class for alert handling.

5. Appendix

Important WinCC OA specific abbreviations

Acronym	Long form	Meaning
WinCC OA	Simatic WinCC Open Architecture	A SCADA system for visualizing and operating of processes, production flows, machines and plants in all lines of business. Distributed systems enable any number of stand-alone systems, from 2 to 2048, to be linked via a network. Each subsystem can be configured either as a single-user or multi-user system, redundant or not, in each case.
DPT	Data point type	Object definition (class) of a structured data object as mapping of a real device. Single data points (instances) are derived from the DPT. Therefore, the data point type is a form of template.
DP	Data point	Structured, device-oriented data object as representation of a real device within the control

		system. A data point contains one or more data point elements (process variables).
DPE	Data point element	Single process information within a device-oriented data point. Each DPE corresponds to a value/state. In addition to the value, there are DPE attributes like time stamp, quality information or origin.
GEDI	Graphic Editor GEDI	Graphic Editor. It is used both for drawing of process images ("panels") as well as for designing of symbols, dialogs, and scripting.
PARA	Configuration Tool	Editor for the creation and configuration of data point types, data points, and data point elements as well as their configs.
ASCII	American Standard Code for Information Interchange	Standardized protocol for storage and transfer of characters/text. In WinCC OA, the acronym also refers to the database import/export manager. It is a module to export and import configurations as ASCII files. Mass configuration can therefore be carried out in a spread sheet program (for example MS Excel), file editor, or in an external database.
D	Driver Manager ("Driver")	Interface for connecting controllers (PLC, DDC, ...) fieldbuses and telecontrol systems. A driver handles the communication via an external protocol and enables the exchange of information with WinCC OA. The processing of data from the “field” to WinCC OA contains event orientation, old/new comparison, transformation, conversion, and smoothing. The protocol of the Driver must be the same as the protocol of the “field” device. Furthermore, the connection (how to reach the device) must be configured in WinCC OA. For exchanging data, a periphery address must be configured on a corresponding DPE.
CTRL	Control Manager ("Scripting")	Processing unit that allows to process user specific logic / business logic (control scripts). Control possesses an easy to learn syntax (similar to ANSI-C) and is processed by an interpreter (CTRL Manager).

Table 4-1 WinCC OA specific abbreviations

Service and support

WinCC OA Extended Services

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Our range of services includes the following:

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- Project startup workshop
- Architecture definition
- Project engineering assistance with dedicated contact person
- Special project developments (special requirements, web widgets, gateways, etc.)
- WinCC OA library development assistance
- Project or architecture reviews with report
- Project-specific problem or performance analysis
- Project upgrade (analysis with report, assistance during upgrade, etc.)
- Assistance for complex error reproduction scenarios
- On-site assistance for any tasks related to WinCC OA and their components
- 24/7 on-duty assistance or priority callback for certain time range
- Database support Oracle®, InfluxDB®, PostgreSQL® and MS SQL®
- Raima/HDB to SQLite/NGA migration
- Setup and consulting for WinCC OA Add-ons e.g., APM, AMS, DRS, …
- WinCC OA Security services (as per the WinCC OA Security Guideline, NIS2)
- Tests on unsupported platforms (e.g., unsupported OS)
- Individual Workshops (driver workshop, UI workshop, business logic, etc.)
- Factory Acceptance Test (FAT)/Site Acceptance Test (SAT) assistance
- Creating of prototypes, proof of concepts, demos, etc.
- Project tender analysis and evaluation of projects

  You can find detailed information on our range of services in the service catalog web page:

www.winccoa.com/documentation/WinCCOA/latest/en_US/Support/topics/support_extendedServices.html

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Change documentation

Version	Date	Modifications
V1.0	01/2026	First version

Table 4-2 Change Documentation

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